Pipe Bend Thickness Report

ASME B31.3-2018 para. 304.2.1 | Pipe bend under internal pressure

DESIGN EQUATION

ASME B31.3-2018, 304.2.1, (3c)-(3e)
t = P D / [2 (S E W / I + P Y)]; tm = t + c
I_in = [4(R1/D) - 1] / [4(R1/D) - 2]
I_out = [4(R1/D) + 1] / [4(R1/D) + 2]
Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

Project	QA — مثال كوع
Prepared by	Device preview
Design reference	B31.3-2018 QA
Report date	2026-10-06 10:57

01 BEND GEOMETRY

Material / grade	User coefficients - example only
Outside diameter D	114.3 mm
Bend radius to centreline R1	171.45 mm (1.5 D)
Inspection / pipe reference	QA-UT

02 DESIGN INPUTS

Internal design gauge pressure P	20 bar(g) = 2 MPa
Allowable stress S	100 MPa
Design metal temperature	20 C
Stress source	B31.3-2018 A-1/A-1M; S supplied at design temperature
Quality factor E / weld factor W	0.85 / 0.9
Coefficient Y	0.4
Corrosion / erosion allowance	1.5 / 0.5 mm
Mechanical removal allowance	1 mm (before fabrication)

03 CALCULATION TRACE

Use the equations above with the following geometry factors. The intrados factor is above 1 and governs the pressure wall.

Intrados factor I	1.25
Extrados factor I	0.875
Pressure thickness at intrados	1.8435 mm
Pressure thickness at extrados	1.2955 mm
Matching straight pipe, I = 1	1.4787 mm
Bend wall before fabrication	4.8435 mm
Required remaining wall at intrados	3.8435 mm

04 MEASURED INTRADOS WALL

Minimum measured intrados wall	4 mm
Required remaining wall	3.8435 mm
Measured minus required	0.1565 mm

MEETS the internal-pressure wall requirement for these inputs.

05 LIMITS AND REFERENCES

Basis: ASME B31.3-2018, 304.2.1. Verify user-supplied S, E, W, Y and allowances against the code and project specification. This report checks required bend wall; it does not predict forming or geometric thinning. Excluded: miter bends (304.2.3), external pressure, other sustained/occasional loads, fatigue and remaining-life assessment.